

Manual

Complex Order Networks HLS-KAN 8.2

Version 1.1.23232

Last changed on: 15.09.2020

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1 Overview: Complex Order Networks

Purpose

This functions package provides functions used in association with relationships (dependencies) covering all orders. You use this function package when you work with linked orders, which may be required in assembly production.

Integration

The relationships (dependencies) between the orders are typically transferred via the interface from the ERP system.

Features

- The function accounts for the linked relationships (dependencies) between the production orders that play a role in manufacturing an assembly.
- Transfers linked relationships (dependencies) via the HYDRA ERP/ PPS interface (requires: SIS-ERP)
- Graphic presentation showing the linked relationships (dependencies)
- Accounts for linked relationships (dependencies) in the event of postponed dates, rescheduled dates or any other planning activities
- Provides the ability to manually define linked relationships (dependencies)

2 Order Network

Overview

HYDRA menu	Production control → Production preparation → Order network
FEDRA menu	Advanced Process Modeling → Current → Order network
Transaction code	ornet
Function authorization	ornet

Purpose

You use this application if you would like to have an overview of an order network, i.e. a view of linked orders.

Integration

The order network displays order planning results in a Gantt chart.

Requirements

For order networks across orders, you must have defined relationships between operations. The relationships between orders, or more accurately, between operations of related orders, must be transmitted from the ERP system via an interface or

Selection criteria

The application provides the following selection criteria:

Order

The order network for this order can be determined using this field. You can also use wildcards.

When an order number is entered, adjacent orders are selected as well. If the adjacent succeeding orders have predecessors, only their last operation is displayed. If the adjacent preceding orders have successors, only their first operation is displayed.

Project number

If a selection is made by project number, all orders are determined that are assigned to the project number entered. This requires that the project number is defined in the order. You can also use wildcards.

Sales order

If a selection is made by customer sales order, all orders are determined that are assigned to the sales order entered. This requires that the sales order number is defined in the order. You can also use wildcards.

Customer name

If you enter a customer name, all orders are determined that are assigned to the customer name you entered. This requires that the customer name is defined in the order. You can also use wildcards.



Note: Only operations are selected and shown that are at least connected to one other operation and have a planned start or planned end. The application does not display operations of orders including only one stage (i.e. orders with one operation only) that are not connected to an operation of another order. Also operations with an identical planned start and end are not displayed.

Generally, only such orders are integrated that have planning identifiers configured for the order type with either F (detailed planning) or T (scheduling).

Toolbar



Zoom out

Reduces the display.



Zoom in

Enlarges the display.



(Grouped by) sales order

Groups the orders and operations by
sales order → order → operation



(Grouped by) project number

Groups the orders and operations by
project number → order → operation



(Grouped by) order

Groups the orders and operations by
Order → Operation

Detail application

Orders are listed on the *left* in table form. Data can be shown or hidden with groupings . Orders are displayed in groups:

- Project number or customer sales order number (depending on how they are grouped)
 - Order
 - Operation

Displayed at the lowest level are operations with the following information:

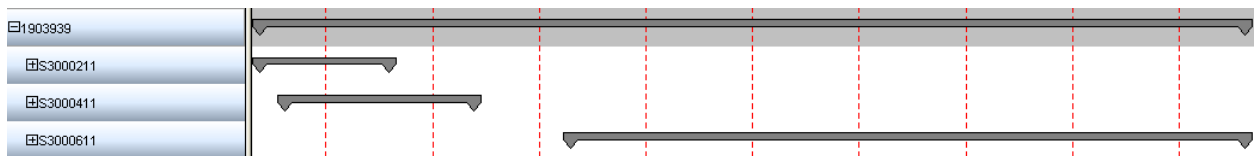
- Operation number
- Workplace at which the operation is planned (if planned)
- Group in which the operation is planned

Operations are shown in the form of a bar on the *right*. For every entry (each line) on the left, there is a corresponding illustration on the right:

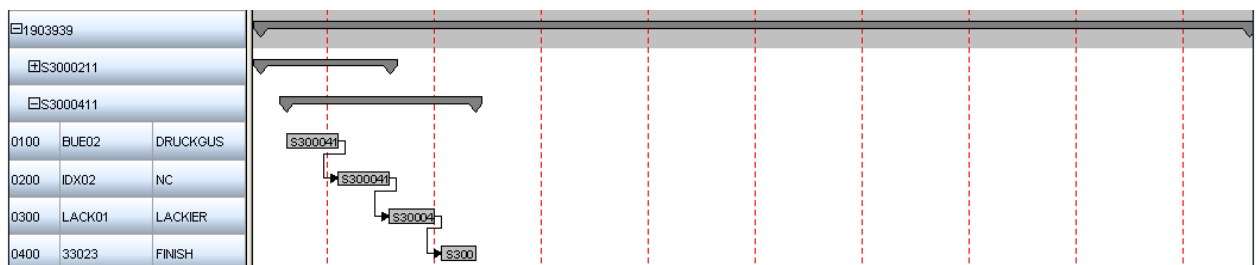
The project order or customer sales order is always displayed in the form of a bar, which is limited at each end with triangles. The length of the bar depends on when each of the orders listed under the project order or sales order, or rather their operations, are scheduled.



If you expand the project order or the sales order, you will see each separate order. Likewise for each order, you will see a bar that is also limited with triangles. The length of the bar depends on when each of the individual operations is scheduled.



When you now expand an order, each of the order's operations is displayed in the form of a bar.



What can now be seen relatively clearly here: the length of the order bar depends on the time domain in which the operations are planned.



All orders are expanded when data is requested and when groups are modified.



The format of the date values displayed on the Gantt chart depends on the format specified by the operating system. The client format is not relevant.

No shift-free times are displayed in the application. The length of the bars in each of the operations is based on the Gregorian calendar. It always equals the length of time between the planned start and the planned end. The coloring of the OP bars is based on the order status configuration.

When you move your mouse cursor over an operation bar, a tool tip is shown that includes the following information:

- MES order number
- Article
- Article designation
- OP designation: operation designation
- Target quantity (P): target quantity in primary quantity unit
- Start: planned start according to planning; if the operation is still in the pool of groups, this is the scheduled start
- End: planned end according to planning; if the operation is still in the pool of groups, this is the scheduled end.

The following applications can be called up from the operation's context menu:

- Order information (function authorization: orin)
- Order progress / order overview (function authorization: orov)
- Edit order network (function authorization: ednet)

Request from graphic planning board

If the order network is called from the graphic planning board, the operations displayed in the Gantt chart are updated with the planned dates scheduled in the graphic planning board, because these may differ from those in the database.

3 Edit Order Network

Summary

HYDRA Menu	Order Management → Order management → Edit order network
FEDRA menu	Advanced Process Modeling → Edit → Edit order network
Transaction code	ednet
Function authorization	ednet

Usage

You use this application to create dependencies for orders beyond the existing operation sequence. These dependencies are referred to as relationships.

Keep in mind that only the end-start relationships can be created. These are relevant for both planning and for data entry. Enter the MES order number (combined order/ OP number) during data entry.

Requirement

The linked orders, including all of their operations, must exist in the system.

Selection criteria

The application provides the following selection criteria:

Order

The relationships are displayed for the selected order number.

OP

The relationships are displayed for the selected operation.

Predecessor/ successor/ predecessor and successor

Only the relationships relating to the selection are displayed.

Toolbar

This application only allows relationships to be created or deleted.



Any relationships created by the system automatically (origin = "S") may not be deleted by the user.

Field descriptions

Predecessor

Order number of the preceding operation

Preceding OP

Operation number of the preceding operation

Successor

Order number of the succeeding operation

Succeeding OP

Operation number of the succeeding operation

Relationship

Only the end-start relationships ("ES") can be created in the setup process.

Origin

Relationships created manually or explicitly via the interface are created using "E" = externally created.

The relationships created by the system are marked with "S".

Active

In principle, relationships are always active. Relationships created due to alternative sequences are the exception. Relationships of inactive alternative sequences are marked as inactive.

Relevance

The system differentiates between relationships for planning and relationships for data entry.

P Relationship is only relevant for planning.

V Relationship is only relevant for data entry.

X Relationship is neither relevant for planning nor for data entry.

<empty> Relationship is relevant for planning and for data entry.

Explicitly set relationships can only be created with relevance =<empty>.